PAPER_1552_Z_RECOMB_1090

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PAPER_1552 — Recombination Redshift = z_recomb = A_5·SO_5 + A_5·D_phys + SO_5·D_crit − SO_5 = 600 + 240 + 260 − 10 = 1090

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Cosmology
Date: June 18, 2026
Status: CLOSED — EXACT closure from PAPER_1209GG_S651

Observation

PAPER_1209GG_S651 (Cosmology Unified Proof Set) lists **Recombination Redshift** with the integer-primitive expression:

```
Recombination Redshift = z_recomb = A_5·SO_5 + A_5·D_phys + SO_5·D_crit − SO_5
= 600 + 240 + 260 − 10 = 1090
```

Status: **EXACT**.

UQFF Closed Identity

```
z_recomb = A_5·SO_5 + A_5·D_phys + SO_5·D_crit − SO_5 = 600 + 240 + 260 − 10 =
1090 EXACT
```

Physical Interpretation

z_recomb sets the last-scattering surface of the cosmic microwave background — the epoch (~380,000 yr after Big Bang) when electrons first bound to nuclei and photons decoupled.

Four integer products with zero fitted constants:
- A_5·SO_5 = 600 (icosahedral × decimal scale)
- A_5·D_phys = 240 (icosahedral × spacetime dimensions)
- SO_5·D_crit = 260 (decimal × bosonic-string critical dimension)
- −SO_5 = −10 (decimal correction)

Sum = 1090 EXACT — the Planck-2018 best-fit value for z_recomb.

This is one of the cleanest cosmological closures in UQFF: the recombination epoch derives from integer-primitive arithmetic alone.

NOT REPLACEMENT

CODATA/PDG treats Recombination Redshift as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209GG_S651
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "z_recomb_1090"})`
